欠阻尼：

标准二阶，无零点，

$$G_0(s)=\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

阶跃响应 $R(s)=\frac{1}{s}$

$$R(s)\cdot G_0(s)=C(s)$$

$$L^{-1}[C(s)]=L^{-1}\left[\frac{1}{s}\cdot\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}\right]$$

改成并联形式

分母因式分解：$s(s^2+2\zeta\omega_n s+\omega_n^2)$

$$=s(s+\zeta\omega_n+j\omega_d)(s+\zeta\omega_n-j\omega_d)$$

$$\Big\Downarrow$$

$$\omega_d=\omega_n\sqrt{1-\zeta^2}$$

$$\frac{A}{s}+\frac{B}{s+\zeta\omega_n+j\omega_d}+\frac{C}{s+\zeta\omega_n-j\omega_d}$$

不完全分解：$\frac{(s^2+2\zeta\omega_n s+\omega_n^2)-(s^2+2\zeta\omega_n s)}{s(s^2+2\zeta\omega_n s+\omega_n^2)}$

$$=\frac{1}{s}-\frac{s+2\zeta\omega_n}{s^2+2\zeta\omega_n s+\omega_n^2}$$

$$\frac{1}{s}-\frac{s+2\zeta\omega_n}{(s^2+2\zeta\omega_n s+\omega_n^2)}$$

$$=\frac{1}{s}-\frac{s+2\zeta\omega_n}{(s+\zeta\omega_n+j\omega_d)(s+\zeta\omega_n\cancel{\varphi}-j\omega_d)}$$

$$=\frac{1}{s}-\frac{s+2\zeta\omega_n}{(s+\zeta\omega_n)^2-(j\omega_d)^2}$$

$$\omega_d=\omega_n\sqrt{1-\zeta^2}$$

$0<\zeta<1$ 欠阻尼

$$=\frac{1}{s}-\frac{s+\zeta\omega_n}{(s+\zeta\omega_n)^2+\omega_d^2}-\frac{\zeta\omega_n}{(s+\zeta\omega_n)^2+\omega_d^2}$$

$$\frac{\zeta}{\sqrt{1-\zeta^2}}\cdot\frac{\omega_d}{(s+\zeta\omega_n)^2+\omega_d^2}$$

② 查表

$f(t)$	$F(s)$
e^{at}	$\frac{1}{s-a}$
$\sin\omega t$	$\frac{\omega}{s^2+\cancel{\omega^2}\omega^2}$
$\cos\omega t$	$\frac{s}{s^2+\omega^2}$

$$1-e^{-\zeta\omega_n t}\cdot\left(\cos\omega_d t+\frac{\zeta}{\sqrt{1-\zeta^2}}\cdot\sin\omega_d t\right)$$

$$=1-e^{-\zeta\omega_n t}\cdot\underbrace{\sqrt{1^2+\frac{\zeta^2}{1-\zeta^2}}}_{\frac{1}{\sqrt{1-\zeta^2}}}\cdot\sin(\omega_d t+\theta)$$

⑪

$$\frac{1}{s} - \frac{s+2\zeta\omega_n}{(s+\zeta\omega_n+j\omega_d)(s+\zeta\omega_n-j\omega_d)}$$

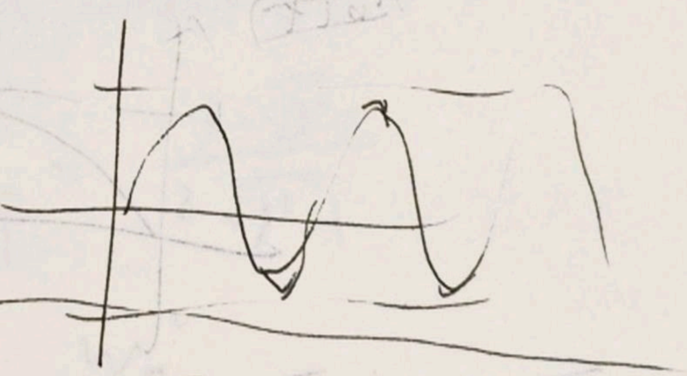
$\zeta = 0$ 无阻尼

$$\frac{1}{s} - \frac{s}{(s+j\omega_d)(s-j\omega_d)}$$

$$= \frac{1}{s} - \frac{s}{s^2+\omega_d^2}$$

$$L^{-1} = 1 - \cos\omega_d t = 1 - \cos\omega_n t$$

$$\overset{\downarrow}{\sqrt{1-\zeta^2}\,\omega_n}$$



$\zeta = 1$ $\qquad \omega_d = \sqrt{1-\zeta^2}\,\omega_n = 0$

$$\frac{1}{s} - \frac{s+2\omega_n}{(s+\omega_n)^2}$$

$f(t) = t \rightarrow F(s) = \frac{1}{s^2}$

$F(s+a) = \frac{1}{(s+a)^2} \rightarrow f(t)e^{-at}$

$$= \frac{1}{s} - \frac{1}{s+\omega_n} - \omega_n \cdot \boxed{\frac{1}{(s+\omega_n)^2}}$$

$$L^{-1} = 1 - e^{-\omega_n}t - \omega_n \cdot t \cdot e^{-\omega_n t}$$

$$= 1 - (1+\omega_n t)\cdot e^{-\omega_n t}$$

续上：$\xi=1$

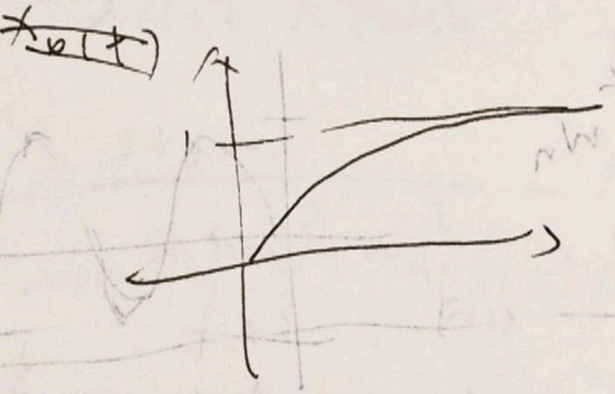
$$x_o(t)=1-(1+\omega_n t)\cdot e^{-\omega_n t}$$

$$\dot{x}_o(t)=-e^{-\omega_n t}(-\omega_n)-[\omega_n\cdot e^{-\omega_n t}+\omega_n t\cdot e^{-\omega_n t}(-\omega_n$$

$$=\omega_n^2 t\cdot e^{-\omega_n t}>0$$ 单升

$t\to 0$ $x_o(t)\to 0$ $x(t)\to 0$

$t\to\infty$ $e^{-\omega_n t}\to 0$ $\dot{x}_o(t)\to 0$ $x(\infty)\to 1$

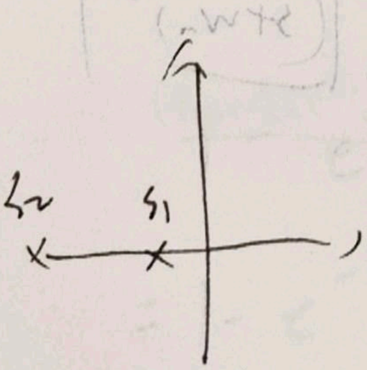


过阻尼 $\xi>1$

$$s^2+2\xi\omega_n s+\omega_n^2=0$$

$$s_{1,2}=-\xi\omega_n\pm\omega_n\sqrt{\xi^2-1}$$

$\xi>1$ 两个负实极点



$$\frac{A}{s} + \frac{B}{s+\zeta\omega_n+\omega_n\sqrt{\zeta^2-1}} + \frac{C}{s+\zeta\omega_n-\omega_n\sqrt{\zeta^2-1}}$$

$$A: \lim_{s\to 0} s\,G(s) = \lim_{s\to 0} s\cdot\frac{\omega_n^2}{s(s+\zeta\omega_n+\omega_n\sqrt{\zeta^2-1})(s+\zeta\omega_n-\omega_n\sqrt{\zeta}}$$

$$= 1$$

$$B: \lim_{s\to -\zeta\omega_n-\omega_n\sqrt{\zeta^2-1}} (s+\zeta\omega_n+\omega_n\sqrt{\zeta^2-1})\frac{\omega_n^2}{s(s+\ \ +)(s+\ \ -)}$$

$$= \frac{\omega_n^2}{(-\zeta\omega_n-\omega_n\sqrt{\zeta^2-1})(-2\omega_n\sqrt{\zeta^2-1})}$$

$$= \frac{1}{2(\zeta+\sqrt{\zeta^2-1})\sqrt{\zeta^2-1}} = \frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}$$

$$C: \lim_{s\to-\zeta\omega_n+\omega_n\sqrt{\zeta^2-1}} \frac{\omega_n^2}{(-\zeta\omega_n+\omega_n\sqrt{\zeta^2-1})(2\omega_n\sqrt{\zeta^2-1})}$$

$$= \frac{1}{2(-\zeta+\sqrt{\zeta^2-1})\sqrt{\zeta^2-1}} = \frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}(-\zeta+\sqrt{\zeta^2-1})(\zeta+\sqrt{\zeta^2-1})}$$

$$= -\frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}$$

$$L^{-1} = 1 + \frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}\cdot e^{-\omega_n(\zeta+\sqrt{\zeta^2-1})t} - \frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}} e^{(-\zeta+\sqrt{\zeta^2-1})\omega_n t}$$

(120) 更负
衰减快

长期间占主导